

Candidate Name : _____

CT Group : _____

Index no : _____

PIONEER JUNIOR COLLEGE
JC2 Preliminary Examination

MATHEMATICS
Higher 2



9740/2

Friday

12th September 2008

Additional materials: Answer Papers
List MF15

TIME 3 hours

INSTRUCTIONS TO CANDIDATES

Do not open this booklet until you are told to do so.

Write your full name, index number and CT group on all the work you hand in.

Write in dark blue or black pen on both sides of the papers.

You may use a soft pencil for any diagrams or graphs.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

Give non-exact numerical answers correct to 3 significant figures or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use a graphic calculator.

Unsupported answers from a graphic calculator are allowed unless a question specifically states otherwise.

Where unsupported answers from a graphic calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers.

The number of marks is given in brackets [] at the end of each question or part question.

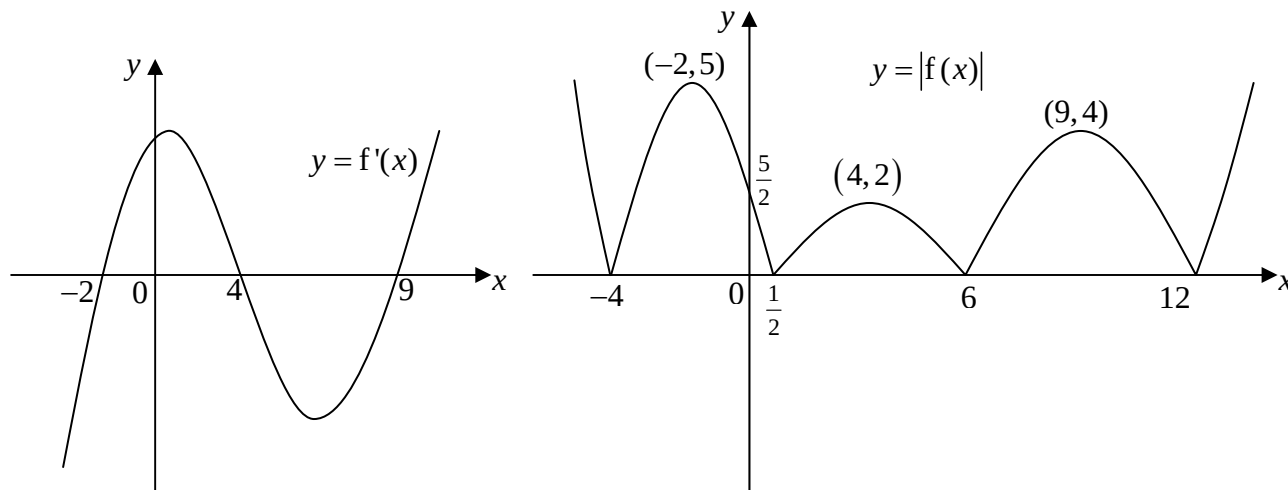
At the end of the examination, arrange your solutions in numerical order and fasten them securely together with the question paper on the top.

For Examiner's Use					
Qn	Marks	Qn	Marks	Qn	Marks
1		6		11	
2		7		12	
3		8		13	
4		9			
5		10			
				Total	

This question paper consists of 6 printed pages.

Section A: Pure Mathematics [40 marks]

- 1 The given sketches show the graphs of $y = f'(x)$ and $y = |f(x)|$.



Sketch the graph of $y = f(x)$, showing clearly the stationary points and intercepts. [3]

- 2 Given that $\sqrt{3} \cos x + \sin x = R \cos(x - \alpha)$ where $0 < \alpha < \frac{\pi}{2}$, show that $R = 2$ and find the exact value of α . [2]

Hence, state precisely a sequence of geometrical transformations which transforms the graph of $y = \cos x$ to $y = 2\sqrt{3} \cos 2x + 2 \sin 2x - 5$. [3]

- 3 The functions f and g are defined by

$$f : x \mapsto 4x - x^2, \quad x \in \mathbb{R}, x \leq A$$

$$g : x \mapsto \frac{x+1}{x-1}, \quad x \in \mathbb{R}, x > 1$$

- (i) Find the largest value of A for which f^{-1} exists and hence find f^{-1} . [3]
- (ii) Show that $f^{-1}g$ does not exist. Hence, find the largest possible domain of g such that $f^{-1}g$ exist. [3]
- 4 (i) By using the substitution $u = \sqrt{2x+1}$, find $\int \frac{e^{\sqrt{2x+1}}}{\sqrt{2x+1}} dx$. [2]
- (ii) Hence, evaluate $\int_0^2 e^{\sqrt{2x+1}} dx$, giving your answer in exact form. [4]

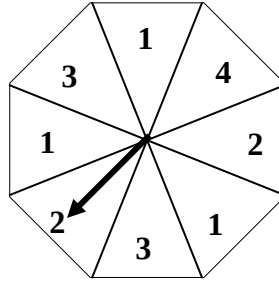
- 5 (a) Write down the modulus and argument of the complex number $w = -1 + i$. Hence find the set of values of negative integer n for which w^n is imaginary. [4]
- (b) Sketch on a single Argand diagram, the loci given by
- (i) $|w + 1 - 8i| = |w - 1 - 6i|$, and [1]
- (ii) $-\frac{\pi}{2} < \arg(z - 2 - 5i) \leq \frac{\pi}{6}$. [3]
- Hence find the range of values of $|z - w|$. [1]
- 6 In a quadrilateral $OPQR$, $\overrightarrow{OP} = -2\mathbf{i} + 8\mathbf{j} + 6\mathbf{k}$, $\overrightarrow{OQ} = 2\mathbf{i} + 5\mathbf{j} + 7\mathbf{k}$ and $\overrightarrow{OR} = 2\mathbf{i} + \mathbf{j} + 11\mathbf{k}$. A is the midpoint of OP and B lies on QR such that the ratio of $QB : BR$ is $1 : 3$.
- (i) Show that the shortest distance from B to the line OP is $\sqrt{\frac{370}{13}}$. [4]
- (ii) Find the value of the angle of ABR . [3]
- (iii) The line L_1 passes through A and is parallel to the vector $-3\mathbf{i} + c\mathbf{j} - \frac{15}{2}\mathbf{k}$. The line L_2 passes through Q and is parallel to the vector $2\mathbf{i} + 3\mathbf{j} + \frac{3}{2}\mathbf{k}$. Show that $c = 6$ when L_1 and L_2 intersect. Hence, find the coordinates of the point of intersection. [4]

Section B: Statistics [60 marks]

- 7 There are 11 hospitals in Singapore that offer stomach banding surgery. To investigate the effectiveness of stomach banding surgery in Singapore, a postgraduate student decides to conduct a survey with 100 patients who had undergone stomach banding surgery at National University Hospital. Explain why this sample is biased. [1]
- It is known that the number of patients who had undergone stomach banding surgery in each hospital is different. Describe how the postgraduate student should obtain a more representative sample of 100 patients in order to determine the effectiveness of stomach banding surgery in Singapore. [3]
- 8 Two families are invited to have a focus group discussion with three male and three female teachers in a school. Each family consists of one student and both his parents. The group takes their places at a round conference table. Find the number of possible arrangements if
- (i) each student must be seated between his parents and the three female teachers must not be seated together, [3]
- (ii) the seats are numbered and the three male teachers must sit together. [2]
- 9 The Consumers Association of Singapore (CASE) investigated the number of complaints made against Xing Xing supermarket and Rong Rong supermarket. The annual number of complaints received against Xing Xing and Rong Rong supermarkets were found to follow a Poisson distribution with mean μ and 8.7 respectively. Given that the probability of no complaints being received against Xing Xing supermarket in three months is equal to the probability of three complaints being received against Rong Rong supermarket in a year, find the value of μ . [3]
- CASE keeps the records of the annual number of complaints against both supermarkets for a long time. Find the probability that the mean number of complaints against Xing Xing supermarket for a random sample of 53 years exceeds twice the mean number of complaints against Rong Rong supermarket for a random sample of 70 years. [4]
- 10 The travelling time for a randomly chosen ferry trip from Mersing to Pulau Tioman may be assumed to have a normal distribution with mean 90 minutes and standard deviation 3.6 minutes.
- A ferry trip is considered to be well-timed if its travelling time differs from the mean travelling time by at most 5 minutes. Show that the probability of a randomly chosen ferry trip being well-timed is 0.835, correct to 3 significant figures. [2]
- A total of 10 ferry trips are made in a day. Assuming independence between the ferry trips, find the least possible value of m such that the probability of at most m trips being well-timed in a day exceeds 0.8. [3]
- A random sample of 55 ferry trips is taken. By means of a suitable approximation, find the probability that less than 10 trips are not well-timed. [4]

[Turn over]

- 11 At a carnival, a game is designed such that each player has 3 spins on a regular octagon turntable as shown in the diagram below.



Each sector has equal area and the scores are indicated on each sector.

- (i) Show that the probability of a player scoring a total of exactly 5 points is $\frac{45}{256}$. [2]
- (ii) Given that the player scores a total of less than 6 points, find the probability that he spins “2” at least once. [4]

On the first and the second days of the carnival, 15 and 43 people played the game respectively. The random variable Y is the total number of players scoring at least 4 points over the two days. State the distribution of Y . [1]

Hence, by using a suitable approximation, find the probability that, over the first two days of the carnival, the total number of player scoring at least 4 points is less than 50. [4]

- 12 The Ministry of Health claims that the average weight of new born babies in Singapore is 3.15kg. An independent study took a random sample of 80 babies. The following results were obtained.

$$\sum x = 256.8, \quad \sum (x - 3.21)^2 = 23.519$$

- (i) Calculate the unbiased estimates of the population mean and variance. [2]
- (ii) Carry out an appropriate test at 7% level of significance to find out if the claim by the Ministry of Health is accurate. [4]
- (iii) State, giving a reason, whether any assumptions about the population are needed in order for the test to be valid. [2]

Given that the null hypothesis is rejected in favour of the alternative hypothesis when the Ministry of Health issues another statement claiming that the average weight of new born babies in Singapore is μ_0 , find the range of possible values of μ_0 . [4]

- 13** Researcher *A* conducts a study comparing the amount of time a 13-year-old spends playing computer games and the increase in myopia. The average number of hours per week the teenager spends playing computer games and the annual increase in myopia of seven 13-year-olds are given below.

Time spent playing computer games, x (no. of hours per week)	7	12	34	25	43	57	66
Annual increase in myopia, y (Degree per year)	25	k	75	50	100	150	200

- (i) State, in terms of k , the coordinates of the point which both the least squares regression lines of y on x and x on y pass through. [2]
- (ii) Given that the least squares regression line of y on x is $y = -12.6 + 2.92x$, find, correct to the nearest integer, the value of k . [2]

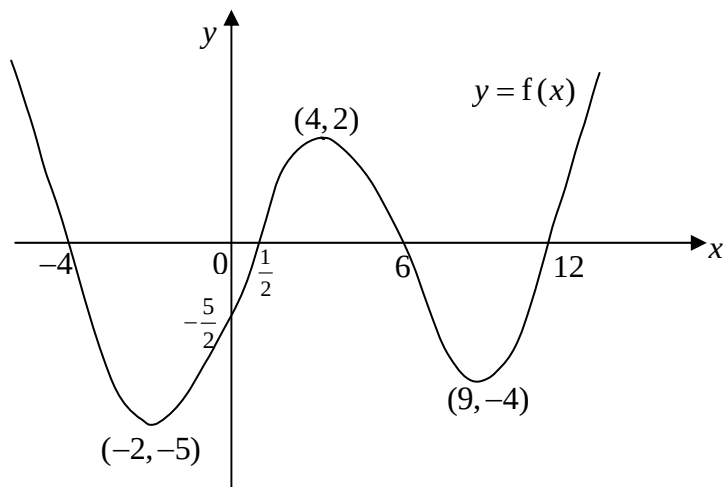
Researcher *B* observes from the scatter diagram that the regression line obtained by researcher *A* may not be suitable.

- (iii) Plot a scatter diagram of y against x using the integer value of k found in (ii) and find the value of the linear product moment correlation coefficient. Hence suggest a suitable transformation, $z = f(y)$, that researcher *B* can use. [3]
- (iv) Find the equation of the least squares regression line of z on x and calculate the value of the linear product moment correlation coefficient. [3]
- (v) Both researchers *A* and *B* estimated the annual increase in myopia of a 13-year-old who spends 50 hours per week playing computer games. Calculate the values obtained by both researchers and comment on the accuracy of the values obtained by researcher *A* and researcher *B*. [2]

End of Paper

Solution

1.



2. Let $\sqrt{3} \cos x + \sin x = R \cos(x - \alpha) = R \cos x \cos \alpha + R \sin x \sin \alpha$
Equating coefficients of $\cos x$ and $\sin x$:

$$\sqrt{3} = R \cos \alpha \quad \dots(1)$$

$$1 = R \sin \alpha \quad \dots(2)$$

Solving (1) and (2), $R = 2$ (shown)

$$\text{and } \alpha = \frac{\pi}{6}$$

$$\therefore \sqrt{3} \cos x + \sin x = 2 \cos\left(x - \frac{\pi}{6}\right)$$

$$\begin{aligned} y &= 2\sqrt{3} \cos 2x + 2 \sin 2x - 5 \\ &= 2\left(\sqrt{3} \cos 2x + \sin 2x\right) - 5 \\ &= 2\left(2 \cos\left(2x - \frac{\pi}{6}\right)\right) - 5 \\ &= 4 \cos\left(2x - \frac{\pi}{6}\right) - 5 \end{aligned}$$

The sequence of transformation:

1. Translation by $\frac{\pi}{6}$ units in the direction of x -axis.
2. Scaling parallel to the x -axis with a factor of $\frac{1}{2}$.
3. Scaling parallel to the y -axis with a factor of 4.
4. Translation by -5 unit in the direction of y -axis.

3. (i) Largest value of $A = 2$

$$\text{Let } 4x - x^2 = y \Rightarrow x^2 - 4x + y = 0$$

By completing the square or formula, $x = 2 \pm \sqrt{4 - y}$

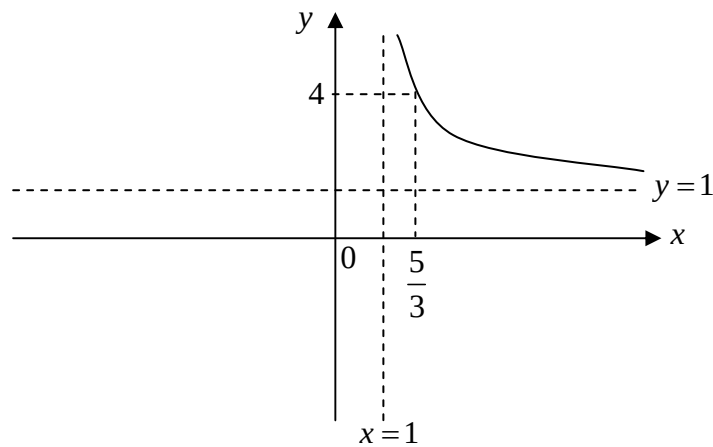
Since $x \leq 2$, reject (+), $\therefore x = 2 - \sqrt{4 - y}$

$$\therefore f^{-1}(x) = 2 - \sqrt{4 - x}, \quad x \leq 4$$

(ii) $R_g = (1, \infty)$, $D_{f^{-1}} = (-\infty, 4]$. Since, $R_g \not\subset D_{f^{-1}}$, therefore $f^{-1}g$ does not exist.

For $f^{-1}g$ to exist, largest possible $R_g = D_{f^{-1}} = (1, 4]$.

From the graph of $y = g(x) = \frac{x+1}{x-1}$,



the corresponding largest possible $D_g = \left[\frac{5}{3}, \infty\right)$.

4. (i) $u^2 = 2x + 1 \Rightarrow 2 \frac{dx}{du} = 2u \Rightarrow dx = u du$

$$\int \frac{e^u}{u} u du = \int e^u du = e^u + C = e^{\sqrt{2x+1}} + C.$$

(ii) $e^{\sqrt{2x+1}} = \frac{e^{\sqrt{2x+1}}}{\sqrt{2x+1}} \sqrt{2x+1}$. Let $u = \sqrt{2x+1}$ $\frac{dv}{dx} = \frac{e^{\sqrt{2x+1}}}{\sqrt{2x+1}}$

Therefore, $\frac{du}{dx} = \frac{1}{\sqrt{2x+1}}$ $v = e^{\sqrt{2x+1}}$

Using by parts,

$$\begin{aligned} \int_0^2 e^{\sqrt{2x+1}} dx &= \left[\sqrt{2x+1} \left(e^{\sqrt{2x+1}} \right) - \int \left(\frac{1}{\sqrt{2x+1}} e^{\sqrt{2x+1}} \right) dx \right]_0^2 \\ &= \left[\sqrt{2x+1} \left(e^{\sqrt{2x+1}} \right) - e^{\sqrt{2x+1}} \right]_0^2 = (\sqrt{5} - 1) e^{\sqrt{5}} \end{aligned}$$

5. (a) $|w| = \sqrt{2} \quad \arg(w) = \frac{3\pi}{4}$

$$w = \sqrt{2} \left(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4} \right)$$

$$w^n = (\sqrt{2})^n \left(\cos \frac{3n\pi}{4} + i \sin \frac{3n\pi}{4} \right)$$

For w^n to be imaginary, $\cos \frac{3n\pi}{4} = 0$

$$\frac{3n\pi}{4} = -\frac{\pi}{2}, -\frac{3\pi}{2}, -\frac{5\pi}{2}, \dots$$

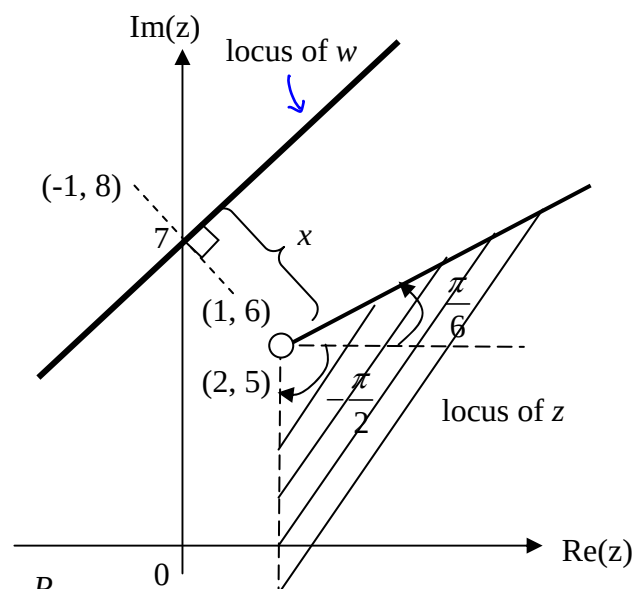
$$n = \frac{2(-1)}{3}, \frac{2(-3)}{3}, \frac{2(-5)}{3}, \dots$$

$$n = \frac{2}{3}(1-2k), \quad k \in \mathbb{Z}^+$$

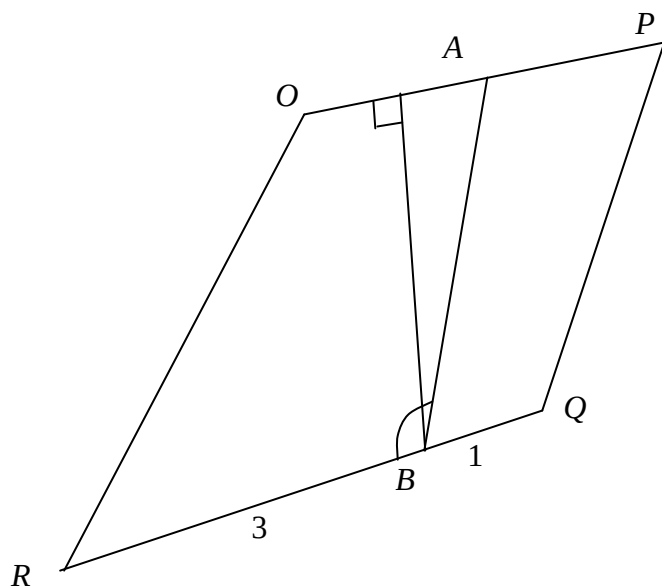
(b) Loci of $|w+1-8i| = |w-1-6i|$ is a perpendicular bisector of the line segment joining $(-1, 8)$ and $(1, 6)$.

$$x = \sqrt{2^2 + 2^2} = \sqrt{8}$$

$\therefore$ Possible range is $|z-w| > \sqrt{8}$



6.



(i) Using Ratio Theorem, $\overrightarrow{OB} = \frac{\overrightarrow{OR} + 3\overrightarrow{OQ}}{4}$

$$= \frac{\begin{pmatrix} 2 \\ 1 \\ 11 \end{pmatrix} + 3\begin{pmatrix} 2 \\ 5 \\ 7 \end{pmatrix}}{4} = \begin{pmatrix} 2 \\ 4 \\ 8 \end{pmatrix}$$

Shortest distance from B to line $OP = \left| \overrightarrow{OB} \times \frac{\overrightarrow{OP}}{|\overrightarrow{OP}|} \right|$

$$= \left| \begin{pmatrix} 2 \\ 4 \\ 8 \end{pmatrix} \times \frac{1}{\sqrt{104}} \begin{pmatrix} -2 \\ 8 \\ 6 \end{pmatrix} \right|$$

$$= \frac{1}{\sqrt{104}} \left| \begin{pmatrix} -40 \\ -28 \\ 24 \end{pmatrix} \right|$$

$$= \sqrt{\frac{370}{13}} \text{ (shown)}$$

(ii) $\overrightarrow{AB} = \overrightarrow{OB} - \frac{1}{2}\overrightarrow{OP} = \begin{pmatrix} 2 \\ 4 \\ 8 \end{pmatrix} - \begin{pmatrix} -1 \\ 4 \\ 3 \end{pmatrix} = \begin{pmatrix} 3 \\ 0 \\ 5 \end{pmatrix}$ $\overrightarrow{RB} = \begin{pmatrix} 2 \\ 4 \\ 8 \end{pmatrix} - \begin{pmatrix} 2 \\ 1 \\ 11 \end{pmatrix} = \begin{pmatrix} 0 \\ 3 \\ -3 \end{pmatrix}$

$$\overrightarrow{AB} \cdot \overrightarrow{RB} = |\overrightarrow{AB}| |\overrightarrow{RB}| \cos \angle ABR$$

$$\begin{pmatrix} 3 \\ 0 \\ 5 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 3 \\ -3 \end{pmatrix} = \sqrt{34} \sqrt{18} \cos \angle ABR$$

$$\angle ABR = \cos^{-1} \left(\frac{-15}{\sqrt{612}} \right) = 127.3^\circ$$

(iii) Eqn of L_1 : $\mathbf{r} = \begin{pmatrix} -1 \\ 4 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} -3 \\ c \\ -\frac{15}{2} \end{pmatrix}, \lambda \in \mathbb{R}$

Eqn of L_2 : $\mathbf{r} = \begin{pmatrix} 2 \\ 5 \\ 7 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ 3 \\ \frac{3}{2} \end{pmatrix}, \mu \in \mathbb{R}$

For the lines to intersect,

$$-1 - 3\lambda = 2 + 2\mu \quad \text{--- (1)}$$

$$4 + c\lambda = 5 + 3\mu \quad \text{--- (2)}$$

$$3 - \frac{15}{2}\lambda = 7 + \frac{3}{2}\mu \quad \text{--- (3)}$$

Solving equations (1) and (3), $\lambda = -\frac{1}{3}$ and $\mu = -1$.

Sub into (2): $c = 6$ (shown)

Thus, the point of intersection is $\begin{pmatrix} 0 \\ 2 \\ \frac{11}{2} \end{pmatrix}$.

7. National University Hospital (NUH) is not the only hospital in Singapore that offers stomach banding surgery. By taking a sample only from NUH, there is a bias in the selection of the sample as patients from other hospitals are not taken into consideration.

The postgraduate student should use stratified random sampling. By considering the proportion of the total number of stomach banding surgeries conducted in the different hospitals in Singapore, a proportional sample size from each hospital can be calculated. For each hospital, the students can now request for all the patient records of those who have undergone stomach banding and use simple random sampling to select the sample size required from each hospital to conduct his survey.

8. (i) Taking each student and his or her parents as a unit, the number of ways to arrange them and the 3 male teacher in a circle = $(5-1)!$

Each student and parents can arrange themselves in $2!$ ways such that the student is seated between parents.

As the 3 female teachers must be separated, the number of ways to arrange them = 5P_3

Therefore, the total number of arrangements = $(5-1)!(2!)(2!)({}^5P_3) = 5760$

- (ii) Taking the 3 male teachers as a unit and arranging all of them in a straight line (since seats are numbered), the number of ways to arrange all of them = $(10!)(3!)$

If the 3 male teachers are seated at seat number 1,2, 12 or 1, 11, 12, the number of ways to arrange them = $(9!)(3!)(2)$

Therefore, the total number of arrangements = $(10!)(3!) + (9!)(3!)(2) = 26127360$

Alternative:

Number of arrangements in a circle without numbering = $(9!)(3!)$

Since there are 12 people in total, therefore the total number of arrangements = $(9!)(3!)(12) = 26127360$

9. $X \sim$ no. of complaints against Xing Xing supermarket in three months.

$$X \sim \text{Po}\left(\frac{1}{4}\mu\right)$$

$Y \sim$ no. of complaints against Rong Rong supermarket in a year. $Y \sim \text{Po}(8.7)$

Given $P(X = 0) = P(Y = 3)$

$$\Rightarrow \frac{e^{-\frac{1}{4}\mu} \left(\frac{1}{4}\mu\right)^0}{0!} = \text{poissonpdf}(8.7, 3)$$

$$\Rightarrow e^{-\frac{1}{4}\mu} = 0.018283$$

$$\Rightarrow \mu = 16.007 = 16.0 \text{ (3 s.f.)}$$

Since $n = 53$ is large, by CLT, $\bar{X} \sim N\left(16.007, \frac{16.007}{53}\right)$ approx.

Since $n = 70$ is large, by CLT, $\bar{Y} \sim N\left(8.7, \frac{8.7}{70}\right)$ approx.

To find $P(\bar{X} > 2\bar{Y}) = P(\bar{X} - 2\bar{Y} > 0)$

$$E(\bar{X} - 2\bar{Y}) = 16.007 - 2(8.7) = -1.393,$$

$$\text{Var}(\bar{X} - 2\bar{Y}) = \frac{16.007}{53} + (2)^2 \left(\frac{8.7}{70}\right) = 0.79916$$

$$\therefore \bar{X} - 2\bar{Y} \sim N(-1.393, 0.79916)$$

$$P(\bar{X} - 2\bar{Y} > 0) = \text{normalcdf}(0, 10^{99}, -1.393, \sqrt{0.79916}) = 0.0596 \text{ (3 s.f.)}$$

10. $X \sim$ traveling time for a ferry from Mersing to Tioman $X \sim N(90, 3.6^2)$

$$P(85 \leq X \leq 95) = \text{normalcdf}(85, 95, 90, 3.6) = 0.83513 = 0.835 \text{ (3 s.f.)}$$

$Y \sim$ no. of ferry trips that are well-timed out of 10

$$Y \sim B(10, 0.83513)$$

$$P(Y \leq m) > 0.8$$

$$\text{Binomcdf}(10, 0.83513, m) > 0.8$$

Using GC or trial and error,

$$P(Y \leq 8) = 0.5092$$

$$P(Y \leq 9) = 0.83498$$

Least $m = 9$

$W \sim$ no. of ferry trips not well-timed out of 55. $W \sim B(55, 0.16487)$

$n = 55$ is large, $np = 9.0679 > 5$, $n(1-p) = 45.932 > 5$

$\therefore W \sim N(9.0679, 7.5728)$ approx.

$$P(W < 10) = P(W < 9.5) \text{ (c.c.)}$$

$$= \text{normalcdf}(-10^{99}, 9.5, 9.0679, \sqrt{7.5728}) = 0.562 \text{ (3 s.f.)}$$

$$\begin{aligned}
11. \quad (i) \quad & P(\text{scoring exactly 5 points}) = P(\text{spinning 1, 1, 3}) + P(\text{spinning 1, 2, 2}) \\
& = \left(\frac{3}{8}\right)\left(\frac{3}{8}\right)\left(\frac{2}{8}\right)\left(\frac{3!}{2!}\right) + \left(\frac{3}{8}\right)\left(\frac{2}{8}\right)\left(\frac{2}{8}\right)\left(\frac{3!}{2!}\right) \\
& = \frac{45}{256}
\end{aligned}$$

$$\begin{aligned}
(ii) \quad & P(\text{spins "2" at least once | scoring a total of less than 6 points}) \\
& = P\left(\frac{\text{spin "2" at least once and score less than 6 points}}{\text{score less than 6 points}}\right) \\
& = \frac{P(\text{spinning 1, 1, 2}) + P(\text{spinning 1, 2, 2})}{P(\text{total 3 points}) + P(\text{total 4 points}) + P(\text{total 5 points})} \\
& = \frac{\left(\frac{3}{8}\right)\left(\frac{3}{8}\right)\left(\frac{2}{8}\right)\left(\frac{3!}{2!}\right) + \left(\frac{3}{8}\right)\left(\frac{2}{8}\right)\left(\frac{2}{8}\right)\left(\frac{3!}{2!}\right)}{\left(\frac{3}{8}\right)\left(\frac{3}{8}\right)\left(\frac{3}{8}\right) + \left(\frac{3}{8}\right)\left(\frac{3}{8}\right)\left(\frac{2}{8}\right)\left(\frac{3!}{2!}\right) + \frac{45}{256}} \\
& = \frac{\frac{27}{256} + \frac{9}{128}}{\frac{27}{512} + \frac{27}{256} + \frac{45}{256}} \\
& = \frac{10}{19} \text{ (or 0.526 to 3.s.f.)}
\end{aligned}$$

$$\begin{aligned}
P(\text{scoring at least 4 points}) &= 1 - P(\text{scoring 3 points}) \\
&= 1 - \frac{27}{512} \\
&= \frac{485}{512}
\end{aligned}$$

$$X_1 \sim \text{no. of people scoring at least 4 points out of 15.} \quad X_1 \sim B\left(15, \frac{485}{512}\right)$$

$$X_2 \sim \text{no. of people scoring at least 4 points out of 43.} \quad X_2 \sim B\left(43, \frac{485}{512}\right)$$

$$\text{Let } Y = X_1 + X_2 \sim B\left(58, \frac{485}{512}\right).$$

Consider Y' which is the no. of people scoring less than 4 points out of 58.

$$Y' \sim B\left(58, \frac{27}{512}\right)$$

Since $n = 58$ is large, p is small such that $np = (58)\left(\frac{27}{512}\right) = 3.0586 < 5$, $Y' \sim$

$Po(3.0586)$ approx.

$$P(Y < 50) = P(Y' > 8) = 1 - \text{poissoncdf}(3.0586, 8) = 0.00430$$

12. (i) $\bar{x} = \frac{\sum x}{n} = \frac{256.8}{80} = 3.21$

Since $\bar{x} = 3.21$, $\sum (x - 3.21)^2 = \sum (x - \bar{x})^2 = 23.519$ is given.

$$s^2 = \frac{1}{n-1} \sum (x - \bar{x})^2 = \frac{1}{79} (23.519) = 0.29771 = 0.298 \text{ (3 s.f.)}$$

(ii) Test $H_0: \mu = 3.15$ vs $H_1: \mu \neq 3.15$

Level of significance: 7 %

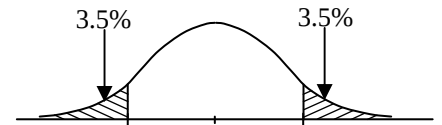
Standardised test statistic: $z = \frac{\bar{x} - \text{"claimed value"}}{\frac{s}{\sqrt{n}}} = 0.984$

From GC, $p\text{-value} = 0.325 > 0.07$

Conclusion: Since the $p\text{-value}$ is more than the level of significance, we do not reject H_0 . There is insufficient evidence, at the 7% level, to indicate that the average weight of new born babies is not 3.15kg.

(iii) No. It is not necessary to for any assumption since sample size is large, hence by Central Limit Theorem, $\bar{X}$ is normally distributed approximately.

Last part: $z = \frac{\bar{x} - \mu_0}{\frac{s}{\sqrt{n}}} = \frac{3.21 - \mu_0}{\sqrt{\frac{0.29771}{80}}} = \frac{3.21 - \mu_0}{0.061003}$



Critical value = $\text{invnorm}(0.965, 0, 1) = 1.8119$

$$\frac{3.21 - \mu_0}{0.061003} < -1.8119 \text{ or } \frac{3.21 - \mu_0}{0.061003} > 1.8119$$

$$3.21 - \mu_0 < -0.11053 \text{ or } 3.21 - \mu_0 > 0.11053$$

$$\mu_0 > 3.32 \quad \text{or} \quad \mu_0 < 3.10$$

13.

- (i) The coordinate of the point that passes through both regression lines is
- $(\bar{x}, \bar{y})$
- .

From GC, $\bar{x} = 34.857 = 34.9$ (3 s.f.)

$$\bar{y} = \frac{\sum y}{n} = \frac{600+k}{7}$$

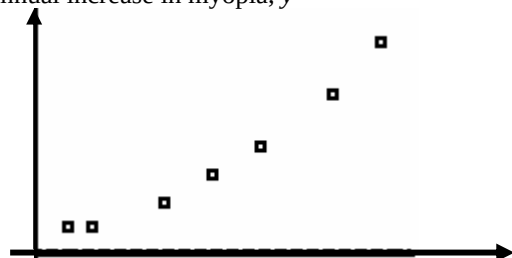
Therefore, point required is $\left(34.9, \frac{600+k}{7}\right)$.

- (ii) Since the regression line of
- y
- on
- x
- passes through
- $\left(34.857, \frac{600+k}{7}\right)$
- ,

$$\frac{600+k}{7} = -12.6 + 2.92(34.857)$$

$$k = 24.277 = 24 \text{ (to the nearest integer)}$$

- (iii) Annual increase in myopia,
- y

No. of hours playing computer games per week, x Linear product moment correlation coefficient $r = 0.978$ Since shape of graph follows the shape of an exponential graph, a suitable transformation is $z = \ln y$ OR $z = \lg y$

(iv)	x	7	12	34	25	43	57	66
	$z = \ln y$	3.2189	3.1781	4.3175	3.912	4.6052	5.0106	5.2983

Linear regression line of z on x : $z = 2.9216 + 0.037252x$
 $\Rightarrow \ln y = 2.92 + 0.0373x$

Linear product moment correlation coefficient $r = 0.991$ OR (if using $z = \lg y$)

x	7	12	34	25	43	57	66
$z = \lg y$	1.3979	1.3802	1.8751	1.699	2	2.1761	2.301

Linear regression line of z on x : $z = 1.2688 + 0.016178x$
 $\Rightarrow \lg y = 1.27 + 0.0162x$

Linear product moment correlation coefficient $r = 0.991$

(v) Researcher *A* obtains $y = -12.6 + 2.92(50) = 133.4 = 133$ (3.s.f.)

Researcher *B* obtains $\ln y = 2.9216 + 0.037252(50) = 4.7842$

$$\Rightarrow y = 120 \text{ (3 s.f.)}$$

OR (if using $z = \lg y$)

Researcher *B* obtains $\lg y = 1.2688 + 0.016178(50) = 2.077$

$$\Rightarrow y = 120 \text{ (3 s.f.)}$$

Researcher *B*'s value is more accurate as the product moment correlation for the model used by *B* is closer to 1 than the product moment correlation for the model used by *A*.